



INTRODUCTION TO BUSINESS ANALYTICS BANA (607)

Prof. Pouyan Eslami

THE IMPACT OF COVID-19 IN LOS ANGELES

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INTRODUCTION:

As of 2024, Los Angeles has faced a major impact from the COVID-19 pandemic. In the early stages, the city experienced a huge number of cases. By mid-2021, Los Angeles had reported over 1.2 million cases and more than 24,000 deaths. The city's hospitals were under intense pressure, especially during the peak of the pandemic, with ICU beds often full. When vaccines became available in 2021, Los Angeles worked hard to get shots to as many people as possible. By the middle of that year, the city had administered over 10 million vaccine doses. Despite challenges like vaccine hesitancy, over 70% of eligible residents received at least one dose by the end of 2021.

The COVID-19 pandemic had an extreme impact on Los Angeles (LA), making it a critical area for studying the different effects of such a crisis on public health, the economy, and communities. We analysed a variety of datasets, including PPP loans, ICU capacity, vaccination rates, and neighbourhood-level covid data, to better understand this. Each of these datasets provided distinct insights into various facets of the pandemic's difficulties and reactions.

The PPP loans dataset provides information about the financial challenges that LA companies faced throughout the pandemic. This dataset uses financial aid distribution to discover the industries and locations that had the most difficulties surviving. It also explains how simple and effective government support programs are for small businesses in high-risk locations.

The ICU capacity data demonstrates the pressure on LA's medical system at the height of the pandemic. It examines the accessibility of essential resources and the responses of various parts of the city to the increase in COVID-19 cases. This data allows us to pinpoint deficiencies in healthcare access and readiness, offering important insights for upcoming public health crises.

The vaccination rates dataset shows how LA's population with diverse ethnicity handled the vaccination programs. Through the analysis of vaccination rates in relation to COVID-19 cases and mortality, it is possible to determine the areas with low vaccination coverage and investigate the possible causes. This data demonstrates the need to break through the social, economic, and cultural barriers to access vaccines.

The neighbourhood-level covid data offers an in-depth perspective on the pandemic's effects across various areas, highlighting the influence of essential social and economic inequalities. Several factors, such as housing density, income levels, and access to healthcare, greatly impacted how different communities faced and managed the crisis. This data emphasizes the need to focus resources on addressing and mitigating structural disparities to build more equitable communities.

By examining these datasets together, we gain meaningful insights into how Los Angeles handled the challenges of the pandemic. The findings highlight the city's areas of strength while also pointing to opportunities for enhancing equity and resilience in future public health emergencies.

RESEARCH QUESTIONS

1. Is there a significant difference between the average of Initial Approval Amount in terms of business that are either part of Hubzone or not?

Dataset: PPP_Loans_LA

Data Analysis:

Variables to be used to perform hypothesis testing for the above research question are Initial Approval Amount and Hubzone indicator.

Statistics for Initial Approval Amount

Mean: 27122.775477394014

Median: 20800.0

Mode: 72188.5

Standard Deviation: 22915.669884977782

The mean (\$27,122.78) for Initial Approval Amount is significantly higher than the median (\$20,800) indicating a right-skewed distribution. Most loans are concentrated around lower values, with some higher-value loans pulling the mean upward. The standard deviation(\$22,915.67) shows that there is high variability in the loan amounts suggesting a wide range of loan sizes.

Summary for HubzoneIndicator

Counts:

HubzoneIndicator

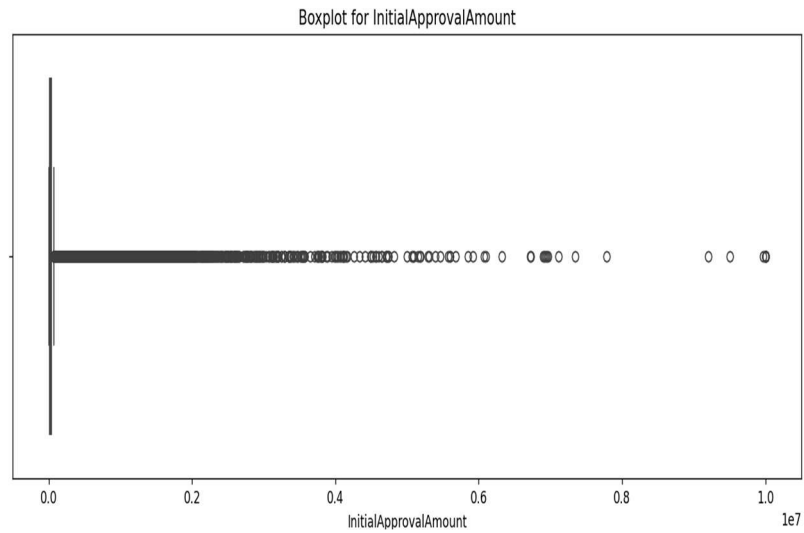
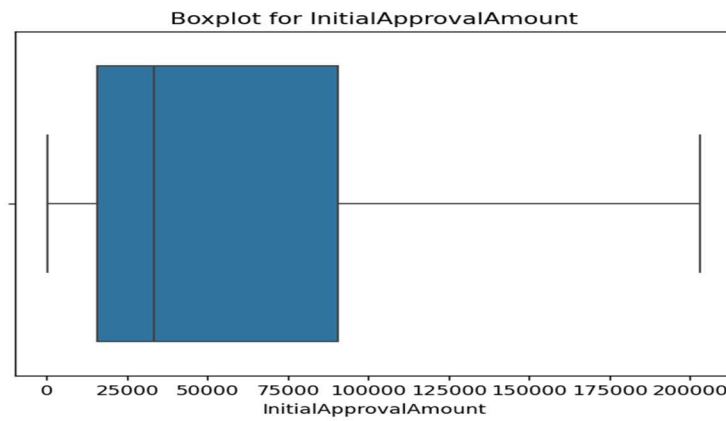
True : 54626

False : 51806

Business in hubzone are 51.32% and 48.68% are not in the hubzone. So, nearly the distribution is equal between businesses in HubZone areas (True) and those outside (False), with a slight majority in HubZones. This distribution suggests that the funding is distributed almost equally to economically disadvantaged areas and other areas indicating equitable allocation.

Outliers:

We have detected an excessive number of outliers for InitialApprovalAmount using the IQR rule. Removing the outliers will result in huge loss of data, So instead of removing the outliers we decided to perform outlier capping in order to normalize the data further for better analysis result.

Before capping:**After capping:**

T-test is used to solve the research question because t-test compares the means of initial approved amount for two groups to determine if they are significantly different from each other and the two groups here are 'TRUE' and 'FALSE' groups of hubzone indicator variables.

Code:

```
# T-test
group_paidinfull = df[df['HubzoneIndicator'] == False]['InitialApprovalAmount'].dropna()
group_exemption4 = df[df['HubzoneIndicator'] == True]['InitialApprovalAmount'].dropna()
# There is a significant difference between the means of these two groups.
t_stat, p_value = stats.ttest_ind(group_paidinfull, group_exemption4)
print(f'T-Test Results: t-statistic = {t_stat}, p-value = {p_value}')
```

Output: T-Test Results: t-statistic = 13.93552997487429, p-value = 4.213598896226299e-8

Finding: Based on the obtained p-value, which indicates the probability that the observed difference is due to chance, there is a difference in terms of mean between businesses that are part of Hubzone which suggests that the HubZone status has a significant impact on the average initial approval amount of PPP loans.

Recommendation: Increased Funding for HubZones: Consider allocating additional resources or setting higher approval caps for businesses in HubZone areas to address their potentially greater financial needs.

RESEARCH QUESTION 2

Is there any significant difference between the variance of ICU capacity percentage across different regions?

Dataset:

Cdph-regional-stay-home-metrics

Data Analysis:

Statistics for icu_capacity_percent

Mean: 0.15116285714285713

Median: 0.13899999999999996

Mode: 0.0

Standard Deviation: 0.12352726364197304

The mean ICU capacity utilization (15.12%) indicates that that usage of ICUs was on moderate level across the observed dataset. The distribution is relatively symmetric as the mean and median (13.90%) are close around this central value. The mode (0.00%) suggests that there are many facilities reporting no ICU capacity utilization. This could indicate reporting gaps or regions without ICU facilities.

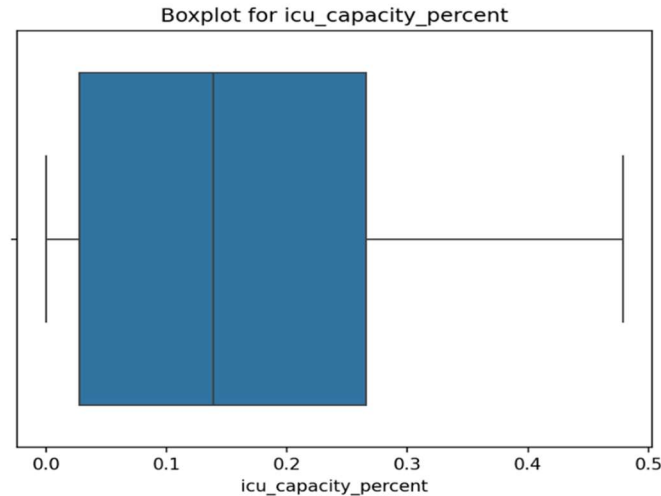
Counts:

Loan Status

Paid in Full 65613

Exemption 40813

The significant majority (61.65%) of businesses were able to repay the loan completely. This high percentage reflects the program's effectiveness in allowing borrowers to pay their debts. 38.35% of them were exempted or forgiven indicating that the PPP loan program helped vast number of businesses to survive during the pandemic.



F-test is used to solve the research question

Code:

```
# Hypothesis Testing
group_Bay_Area = df[df['region'] == 'Bay Area']['icu_capacity_percent'].dropna()
group_Greater_Sacramento = df[df['region'] == 'Greater Sacramento']['icu_capacity_percent'].dropna()
group_Northern_California = df[df['region'] == 'Northern California']['icu_capacity_percent'].dropna()
group_San_Joaquin_Valley = df[df['region'] == 'San Joaquin Valley']['icu_capacity_percent'].dropna()
group_Southern_California = df[df['region'] == 'Southern California']['icu_capacity_percent'].dropna()
f_stat, f_p_value = stats.levene(group_Bay_Area, group_Greater_Sacramento, group_Northern_California, group_San_Joaquin_Valley, group_Southern_California)
print(f'F-Test Results: F-statistic = {f_stat}, p-value = {f_p_value}')
```

Output: F-Test Results: F-statistic = 3.916500930073466, p-value = 0.0040012406462571195

Finding: The test results show that variability in ICU capacity percentages are not similar among the five regions (Bay Area, Greater Sacramento, Northern California, San Joaquin Valley, and Southern California). The low p-value (0.004) indicates a statistically significant difference in the variability of ICU capacity usage across different regions. Some regions may have much higher or lower variability in ICU capacity utilization compared to others, possibly reflecting differences in healthcare infrastructure, population density, or demand for ICU resources.

Recommendation:

Some regions may have much higher or lower variability in ICU capacity utilization compared to others, reflecting differences in healthcare infrastructure, population density, or demand for ICU resources. These results imply that to successfully address requirements, healthcare policy and resource allocations may need to take regional variations into account.

RESEARCH QUESTION 3

Is the capability of paying off the loan associated with the type of business?

Description: This research question is about the association between the variables 'Loan status' and 'Industry' to know if certain types of business could survive better during the pandemic and could repay the loan.

Dataset: PPP_Loans_LA

Data Analysis:

LoanStatus

Paid in Full 65613

Exemption 40813

The significant majority (61.65%) of businesses were able to repay the loan completely. This high percentage reflects the program's effectiveness in allowing borrowers to pay their debts. 38.35% of them were exempted or forgiven indicating that the PPP loan program helped vast number of businesses to survive during the pandemic.

Chi-square test is used to solve the research question. It is used to measure the relationship between loan status and Industry.

Code:

```
# Chi-square test
contingency_table = pd.crosstab(df['Industry'], df['LoanStatus']).dropna()
chi2_stat, chi2_p_val, dof, expected = stats.chi2_contingency(contingency_table)
print(f'Chi-Squared Test Results: chi2_stat = {chi2_stat}, p-value = {chi2_p_val}')
```

Output: Chi-Squared Test Results: chi2_stat = 2768.641395369156, p-value = 0.006

Finding: The test results indicate that the variables loan status and industry are not independent. There is a significant association between the loan status and type of business.

Recommendation: Provide automatic or additional forgiveness for industries with demonstrated difficulty in recovery, such as entertainment, tourism, and small-scale manufacturing. For industries disproportionately impacted (e.g., hospitality, retail), extend repayment schedules or offer temporary interest waivers.

RESEARCH QUESTION 4

What is the relationship between the number of COVID-19 cases, population size, and the number of deaths in neighbourhood-level covid data, and how well can these factors predict mortality rates?

Description

This study aims to evaluate the relationship between the number of deaths and predictors such as the number of COVID-19 cases, population size, and cases per 100,000 individuals. Simple and multiple linear regression models were applied to identify statistically significant predictors and quantify their influence on deaths.

Dataset: Neighbourhood-level COVID-19 data including cases, population, cases per 100,000, and deaths.

OLS Regression Results						
=====						
Dep. Variable:	deaths	R-squared:	0.695			
Model:	OLS	Adj. R-squared:	0.695			
Method:	Least Squares	F-statistic:	1.139e+05			
Date:	Wed, 04 Dec 2024	Prob (F-statistic):	0.00			
Time:	19:18:46	Log-Likelihood:	-2.8266e+05			
No. Observations:	50000	AIC:	5.653e+05			
Df Residuals:	49998	BIC:	5.653e+05			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	33.2577	0.483	68.866	0.000	32.311	34.204
cases	0.0103	3.05e-05	337.523	0.000	0.010	0.010
=====						
Omnibus:	16690.087	Durbin-Watson:	2.001			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	102559.834			
Skew:	1.471	Prob(JB):	0.00			
Kurtosis:	9.370	Cond. No.	2.48e+04			

Methods

1. Simple Linear Regression:

- **Predictor:** Cases
- **Response:** Deaths
- **Purpose:** To measure the direct effect of cases on the number of deaths.

2. Multiple Linear Regression:

- **Predictors:** Cases, Population, Cases per 100,000
- **Response:** Deaths
- **Purpose:** To analyze the combined influence of predictors on deaths and assess statistical significance.

3. Evaluation Metrics:

- Coefficients ($\beta_1, \beta_2, \beta_3$): Quantify the impact of predictors on the response variable.
- **R-squared:** Measures the proportion of variance explained by the model.
- **Confidence Intervals:** Assesses the range within which the true coefficients are likely to fall.
- **p-values:** Determines statistical significance of predictors.

OLS Regression Results						
=====						
Dep. Variable:	deaths	R-squared:	0.752			
Model:	OLS	Adj. R-squared:	0.752			
Method:	Least Squares	F-statistic:	5.064e+04			
Date:	Wed, 04 Dec 2024	Prob (F-statistic):	0.00			
Time:	19:18:46	Log-Likelihood:	-2.7745e+05			
No. Observations:	50000	AIC:	5.549e+05			
Df Residuals:	49996	BIC:	5.549e+05			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	2.1887	1.220	1.794	0.073	-0.203	4.580
cases	0.0070	8.12e-05	86.535	0.000	0.007	0.007
cases_per100k	-8.683e-05	5.61e-05	-1.549	0.121	-0.000	2.31e-05
population	0.0013	1.82e-05	69.613	0.000	0.001	0.001
=====						
Omnibus:	5855.850	Durbin-Watson:	2.010			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	42345.024			
Skew:	0.328	Prob(JB):	0.00			
Kurtosis:	7.460	Cond. No.	3.13e+05			
=====						

Analysis

Simple Linear Regression Results:

- **R-squared:** 0.695, indicating the model explains 69.5% of the variance in deaths based on cases alone.
- **Coefficient for Cases (β_1):** 0.0103
 - Interpretation: For each additional case, the number of deaths increases by approximately 0.0103 on average.
 - **p-value:** Highly significant ($p < 0.05$).

- **Intercept (β_0): 33.26**
 - Represents the baseline number of deaths when the number of cases is zero.

Multiple Linear Regression Results:

- **R-squared:** 0.752, suggesting the model explains 75.2% of the variance in deaths.
- **Coefficients:**
 - **Cases (β_1): 0.0070**
 - **Confidence Interval:** [0.0069, 0.0072]
 - **p-value:** Significant ($p < 0.05$).
 - Interpretation: A positive association between cases and deaths.
 - **Population (β_2): 0.0013**
 - **Confidence Interval:** [0.0012, 0.0013]
 - **p-value:** Significant ($p < 0.05$).
 - Interpretation: Larger populations are associated with more deaths.
 - **Cases per 100,000 (β_3): -8.68e-05**
 - **Confidence Interval:** [-0.0002, 0.00002]
 - **p-value:** Not significant ($p > 0.05$).
 - Interpretation: No meaningful contribution to the number of deaths.
- **Intercept (β_0): 2.1887**
 - Represents the baseline number of deaths when all predictors are zero.

Result:

1. **Model Statistic:**
 - **Significant Predictors:**
 - Cases ($\beta_1=0.0070$): Strong positive association with deaths.
 - Population ($\beta_2=0.0013$): Significant positive influence on deaths.
 - **Non-significant Predictor:**
 - Cases per 100,000 ($\beta_3=-8.68e-05$): Did not significantly contribute to predicting deaths.
2. **Model Performance:**
 - The multiple regression model explained 75.2% of the variance in deaths, an improvement over the simple regression model (69.5%).

3. Practical Implications:

- Reducing the number of cases, especially in densely populated areas, could significantly decrease mortality rates.
- Population size should be a focus in allocating healthcare resources and planning interventions.

4. Limitations:

- Cases per 100,000 did not show significance, which may warrant further exploration with additional covariates or refined data.

Finding:

The analysis of predictors for mortality rates due to COVID-19 reveals several important insights. Notably, the number of COVID-19 cases ($\beta_1=0.0070$) demonstrates a strong positive association with the number of deaths, emphasizing the critical link between case surges and mortality outcomes. This indicates that higher case counts directly contribute to an increase in mortality, potentially due to the strain on healthcare systems, delayed medical responses, or the increased likelihood of severe cases requiring extensive care.

Additionally, population size ($\beta_2=0.0013$) is a significant positive predictor of deaths, underscoring the disproportionate impact of COVID-19 in areas with larger populations. This relationship highlights the importance of tailoring public health strategies to densely populated regions, where the spread of the virus and its consequences may be magnified due to factors like population density and higher rates of human interaction.

Recommendation:

Based on these findings, a two-pronged approach is recommended to mitigate mortality rates effectively. First, targeted efforts to reduce the number of COVID-19 cases are paramount, particularly in densely populated areas. Strategies such as robust testing, contact tracing, and localized lockdowns during outbreaks could significantly curtail the spread of the virus, directly impacting mortality rates. Public health campaigns focusing on vaccine uptake, mask-wearing, and other preventive measures should be prioritized in these high-risk regions.

Research Question 5

Does Covid Case significantly differ if a city is unincorporated compared to cities that have been incorporated?

Description

The study investigates whether the number of COVID cases differs significantly between areas classified as unincorporated (not part of any city) and incorporated cities. The data includes COVID case counts and vaccination information for two groups: cities and unincorporated areas. Descriptive statistics and an independent two-sample t-test were used to analyze the difference.

Key Statistics

Mean: 5820.67 Cases

Median: 3372.0 Cases

Mode: 2388 Cases

Standard Deviation: 6348.06 Cases

Group Distribution:

City: 205 (64.67%)

Unincorporated: 112 (35.33%)

Method

1. Data Cleaning:

- Missing values were removed or imputed with column means for numerical values and the mode for categorical variables.
- Outliers in the "Cases" variable were identified and removed using the interquartile range (IQR) method.

2. T-Test:

- An independent two-sample t-test was performed to compare the mean COVID cases between unincorporated areas and incorporated cities.
- Assumes that both groups are independent and approximately normally distributed.

3. Hypotheses:

- **Null Hypothesis (H_0):** There is no significant difference in COVID case data between unincorporated areas and incorporated cities, considering vaccination data.
- **Alternative Hypothesis (H_1):** There is a significant difference in COVID case data between unincorporated areas and incorporated cities, considering vaccination data.

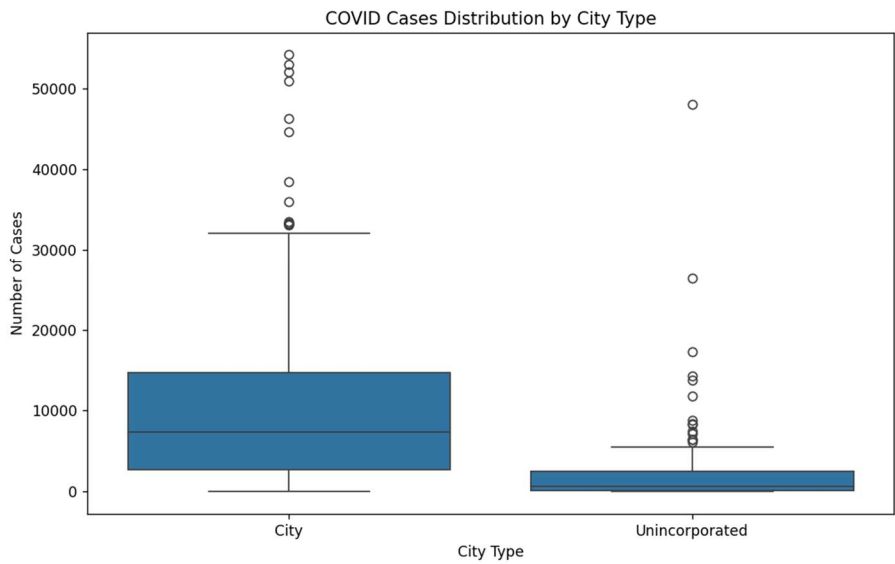
Analysis

1. Descriptive Analysis:

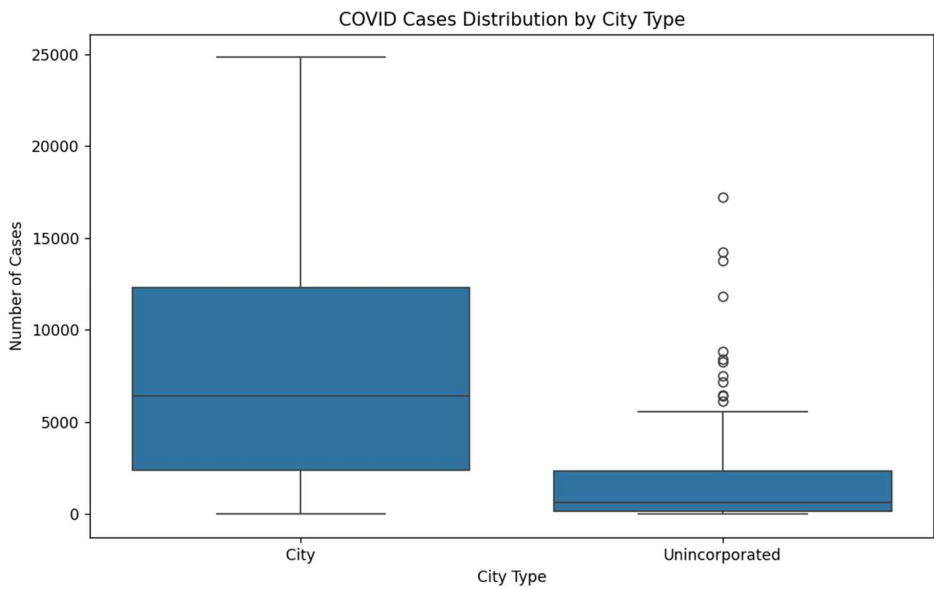
- The data indicates variability in COVID cases, with a high standard deviation (6348.06) suggesting substantial dispersion across the dataset.
- Unincorporated areas make up 35.33% of the dataset, while incorporated cities account for 64.67%.
- Visualization

Since the data is skewed and introduces many outliers, we perform capping data using IQR at 25% level which results in a better normalized data.

Before



After



2. T-Test Results:

- **t-statistic:** 8.92
- **p-value:** 3.76e-17

3. The **p-value** is significantly less than the standard significance level ($\alpha=0.05$), indicating strong evidence against the null hypothesis.

Finding:

The results of the independent two-sample t-test, with a t-statistic of 8.92 and a p-value of 3.76e-17, provide compelling evidence against the null hypothesis. This indicates that the mean number of cases differs substantially between these two types of regions. Given the threshold for statistical significance ($\alpha=0.05$), the p-value underscores a strong level of confidence in rejecting the null hypothesis, highlighting a meaningful disparity in COVID-19 case counts.

The descriptive statistics further illuminate this variability. The overall mean number of cases is 5820.67, but the high standard deviation (6348.06) reveals a considerable range in case numbers across different regions. This dispersion suggests that certain cities or unincorporated areas may experience far higher or lower case counts than the average, pointing to potential localized factors influencing COVID-19 spread.

Moreover, the distribution of data shows that 64.67% of the observations come from incorporated cities, while unincorporated areas represent 35.33%. This imbalance in group sizes, while accounted for in the analysis, might reflect real-world differences in population density, healthcare access, or public health interventions, all of which could contribute to the observed disparities in COVID-19 case numbers.

Recommendation:

To address the disparities in COVID-19 case numbers between incorporated cities and unincorporated areas, a targeted approach to public health intervention is recommended. Policymakers and public health officials should consider the unique challenges and characteristics of unincorporated areas when devising strategies to mitigate the spread of the virus.

1. Strengthen Public Health Infrastructure in Unincorporated Areas

Unincorporated areas may lack the administrative and healthcare infrastructure present in incorporated cities. Investments should focus on establishing or enhancing local healthcare facilities, increasing access to testing and vaccination sites, and ensuring that residents have access to accurate and timely public health information. Mobile health units and partnerships with regional healthcare providers could be particularly effective in reaching underserved populations in these areas.

2. Tailored Outreach and Education Campaigns

Public health messaging should be customized to address the specific needs and characteristics of unincorporated areas. These campaigns could emphasize the importance of preventive measures, such as mask-wearing, social distancing, and vaccination, while also addressing any misinformation or hesitancy that may be prevalent in these communities. Collaboration with local leaders and organizations can help build trust and improve the effectiveness of these campaigns.

CONCLUSION:

This analysis highlights how COVID-19 impacted businesses, healthcare systems, and public health in Los Angeles. Businesses in Hub Zone areas received higher loan amounts, showing they were prioritized, but they still need more targeted support. Recommendations include increasing funding flexibility, offering additional loan forgiveness, and extending repayment options for industries that struggled the most, such as hospitality and tourism. This will help businesses recover faster and sustain operations in the long term.

Healthcare systems showed differences in ICU capacity across regions, suggesting the need to allocate more resources to areas with higher variability, such as mobile ICU units and additional staff. Public health efforts should focus on densely populated areas to reduce COVID-19 cases and deaths by promoting vaccination, testing, and preventive measures like mask-wearing. Strengthening public health infrastructure in underserved regions will ensure better preparedness for future crises and help build resilience across Los Angeles. These actions can create a more equitable recovery and prepare the city to handle future challenges effectively.